

Case Management & Analysis Tool

PREFACE

In today's digital era, managing cases and information efficiently has become a major challenge due to the increasing volume of data and complexity of operations. Traditional methods are often time-consuming and lack proper tracking and analysis.

The project “**Case Management and Analysis Tool**” aims to provide a centralized system for managing, tracking, and analyzing cases in an efficient manner. It helps improve workflow, maintain proper records, and support better decision-making through basic analysis and reporting features.

By integrating case tracking and analytical capabilities, the project provides a reliable solution for improving operational efficiency and ensuring transparency in case handling processes. This system can be useful in various domains such as cybersecurity, governance, and organizational management.

I would like to express my sincere gratitude to my faculty guide, colleagues, and institution for their continuous support and guidance throughout this project.

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Case Management And Analysis tool

1. Project Profile

Project Title	Case Management and Analysis tool.
Objective	Automated Threat analysis integration
OS	Windows 10 / Windows 11, Ubuntu (Linux)
Tool	Cortex, Hive, Ubuntu, ElasticSearch
Developed by	Kugashiya Milan 24084321001 Soni Jay 240843210029
Group No.	11
Submitted to	Department of Computer Science, Ganpat University, Kherva

2. Introduction

The Case Management and Analysis Tool is a software-based solution designed to manage, track, and analyze cases in an organized and efficient manner. In today's fast-growing digital environment, organizations deal with a large number of cases such as incidents, service requests, complaints, and security issues. Handling these cases manually or through unstructured systems can lead to delays, errors, and lack of proper monitoring.

This tool provides a centralized platform where all cases can be created, assigned, updated, and resolved systematically. Each case is recorded with relevant details such as case ID, description, priority, status, and assigned user. This ensures better visibility and accountability throughout the case lifecycle.

In addition to case management, the system includes analysis features that help in identifying trends, recurring issues, and performance metrics. Reports and dashboards can be generated to support decision-making and improve overall workflow efficiency.

The tool also enhances data security and maintains proper documentation, which is important for auditing and compliance purposes. By automating routine processes, it reduces manual workload and minimizes the chances of human error.

Overall, the Case Management and Analysis Tool provides an effective solution for improving case handling, increasing productivity, and ensuring transparency. It can be widely used in areas such as cybersecurity, customer support, IT service management, and governance, risk, and compliance (GRC).

2.1. Overview

The Case Management and Analysis Tool are a centralized system that helps in efficiently managing and analyzing different types of cases within an organization. It provides a structured approach to handle cases from creation to closure, ensuring proper tracking and documentation at every stage.

The system allows users to create new cases, assign them to responsible individuals, update their status, and monitor progress in real time. Each case is stored with important details such as priority, category, and resolution history, which helps in maintaining transparency and accountability.

Additionally, the tool offers basic analytical features such as reports and dashboards to identify trends, measure performance, and support better decision-making. This improves overall workflow efficiency and helps organizations respond to issues more effectively.

Overall, the system simplifies case handling, reduces manual effort, and ensures organized and reliable case management across various domains.

2.2. Background and Motivation

In today's digital environment, organizations handle a large number of cases such as incidents, service requests, complaints, and security-related issues. Traditionally, these cases were managed using manual methods or basic tools, which often resulted in poor tracking, delayed responses, and lack of proper documentation.

With the increasing complexity of operations and data, there is a growing need for an efficient and automated system that can manage cases in a structured way. Organizations require better visibility, faster resolution, and the ability to analyze case data to improve performance and decision-making.

The motivation behind developing the Case Management and Analysis Tool is to overcome these challenges by providing a centralized and user-friendly platform. This system helps in organizing case information, automating workflows, and enabling analysis through reports and dashboards.

By implementing such a tool, organizations can reduce manual effort, minimize errors, and improve overall efficiency. It also supports transparency, accountability, and better resource management, making it an essential solution for modern case handling systems.



2.3 Objective

The main objective of the Case Management and Analysis Tool is to develop a centralized system that can efficiently manage and track cases in an organized manner.

The system aims to reduce manual effort and minimize errors by automating case handling processes. It provides real-time tracking and status updates, ensuring better visibility and control over each case.

Additionally, the tool focuses on maintaining proper documentation and records, which helps in accountability and future reference. It also includes basic analysis features to identify trends and recurring issues, supporting better decision-making.

Overall, the objective is to improve workflow efficiency, enhance productivity, and ensure a smooth and transparent case management process.

2.4. Methodology

The methodology for the project “**Case Management and Analysis Tool**” follows a systematic approach divided into six key phases: Requirement Analysis, System Design, Implementation, Testing & Evaluation, Result Analysis, and Conclusion. Each phase ensures that the system is efficient, reliable, and user-friendly.

1. Requirement Analysis

In this phase, the need for an efficient case management system is identified.

A detailed study is conducted to understand how organizations handle cases such as incidents, complaints, and service requests. The system requirements are defined, including:

- Case creation, assignment, and tracking
- User roles (Admin/User)
- Case attributes (ID, description, priority, status)
- Reporting and analysis features

This phase ensures a clear understanding of both functional and non-functional requirements.

2. System Design

The system architecture is designed to manage cases in a structured and organized way.

The process is divided into two main components:

1. Case Management Module:

This module handles the creation, assignment, updating, and closure of cases. It ensures proper tracking and documentation of each case.

2. Analysis Module:

This module provides reporting and dashboard features to analyze case data, identify trends, and measure performance.

The design focuses on simplicity, scalability, and data security.

3. Implementation

The system is implemented in a modular format for easy development and maintenance.

- The frontend is developed to provide a user-friendly interface for managing cases.
- The backend is developed to handle business logic and data processing.
- The database is structured to store case details securely and efficiently.

4. Testing and Evaluation

Testing is performed to ensure that the system works correctly and meets user requirements.

The evaluation parameters include:

- Functionality: Proper working of all modules
- Usability: Ease of use for users
- Performance: Speed and responsiveness of the system
- Security: Protection of case data

5. Result Analysis

In this phase, the performance of the system is analyzed and compared with traditional manual methods.

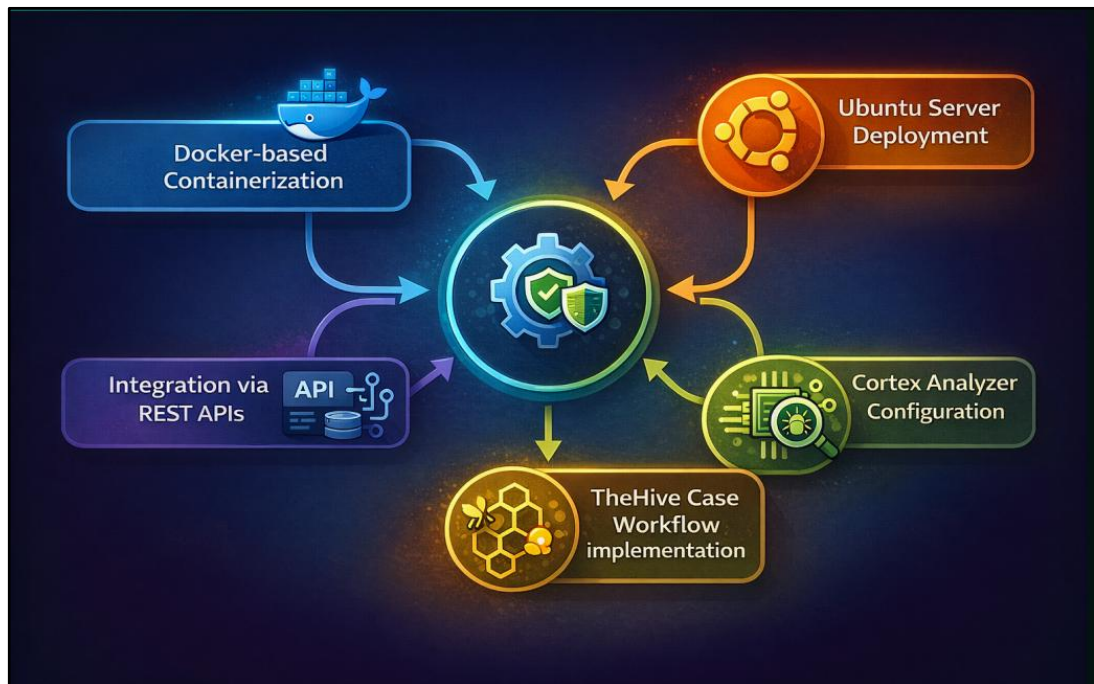
Key observations include:

- Improved efficiency in case handling
- Faster response and resolution time
- Better tracking and documentation
- Enhanced decision-making through reports and analysis

6. Conclusion

The methodology ensures the successful development of a reliable and efficient Case Management and Analysis Tool. The system improves workflow, reduces manual effort, and provides better control over case handling processes.

Case Management And Analysis tool



3. Hardware and Software Requirement

Hardware Requirement:

<u>Component</u>	<u>Specification</u>
Processor	High-Performance Multi-Core Processor
RAM	Minimum 16 GB
Storage	Minimum 100 GB SSD

Software Requirement

<u>Category</u>	<u>Details</u>
Operating System	Ubuntu Server 24.04 LTS
Virtualization	Oracle VM VirtualBox
Containerization	Docker, Docker Compose
Core Tools	TheHive v5, Cortex v3.1.1, Elasticsearch 7.17.9
Analyzers	VirusTotal, MalwareBazaar

4. Functional Specification

Functional Overview

The system performs a structured case handling and analysis process:

Case Management:

Allows users to create, assign, update, and track cases throughout their lifecycle, from initiation to closure.

Case Analysis:

Provides reporting and dashboard features to analyze case data, identify trends, and support better decision-making.

1.Major Functional Modules

a) User Interface Module

- Provides a user-friendly interface for managing cases and viewing reports.
- Allows users to create, update, assign, and track cases easily.

b) Case Management Module

- Handles creation, assignment, updating, and closure of cases.
- Stores details such as case ID, description, priority, and status.

c) User Management Module

- Manages user roles such as Admin and User.
- Controls access and permissions for different system functionalities.

d) Tracking & Monitoring Module

- Tracks the progress and status of each case in real time.
- Ensures transparency and accountability in case handling.

e) Analysis & Reporting Module• Generates reports and dashboards based on case data.

- Helps in identifying trends, performance metrics, and recurring issues.

f) Security & Validation Module

- Validates user authentication and access control.
- Ensures data security and prevents unauthorized access.

2. Input Specifications

- Case details (title, description, priority, category)
- User information (login credentials, roles)
- Case updates and status changes

3. Output Specifications

- Case records with updated status and history
- Reports and dashboards for analysis
- Notifications or alerts for case updates
- Performance summaries and tracking results

5. Tools Description

<u>Layer</u>	<u>Tools</u>	<u>Purpose</u>
Frontend	The Hive Web UI	User interface for case management
Backend	TheHive, Cortex	Business logic and processing
Database	Elasticsearch	Data storage and indexing
Container	Docker	Deployment and orchestration
OS	Ubuntu Server 24.04	Hosting environment
External	VirusTotal, MalwareBazaar	Threat intelligence services

Docker – Containerization Platform

- Used to deploy TheHive, Cortex, and Elasticsearch in isolated containers
- Ensures portability and consistent environment setup
- Simplifies dependency management
- Enables scalability and easy replication of environment
- Reduces system conflicts between services

Ubuntu Server – Operating System

- Base operating system for deployment
- Stable and secure Linux environment
- Optimized for server performance
- Supports containerization and network configuration
- Provides CLI-based system management and security controls

TheHive – Incident Response Platform

Open-source Security Incident Response Platform

- Case creation and alert management
- Task assignment and workflow tracking
- Collaboration between security analysts
- Centralized investigation dashboard

Cortex – Threat Intelligence & Analyzer Engine

- Automated analysis of Indicators of Compromise (IOCs)
- Supports IP, Domain, URL, and File Hash enrichment
- Integration with external threat intelligence sources
- Analyzer execution directly from TheHive cases
- Improves investigation speed and accuracy

REST API – Integration Mechanism

- Enables communication between TheHive and Cortex
- Automates analyzer execution from cases
- Real-time case updates
- Secure API key-based authentication
- Enables future scalability with additional tools

Elasticsearch – Data Storage & Indexing

- Stores alerts, cases, and analysis results
- Provides fast indexing and search capabilities
- Real-time log retrieval
- Scalable data architecture
- Ensures efficient case data management

What is Case Management and Analysis System.

A Case Management and Analysis System is a centralized platform designed to record, manage, analyze, and track cybersecurity incidents or forensic cases.

- It helps organizations organize incidents, assign tasks to analysts, and perform indepth analysis using structured workflows.
- This system improves the efficiency of Security Operations Centers (SOCs) by allowing better visibility, collaboration, and documentation.
- It supports both manual operations (like case creation and task assignment) and automatedanalysis through external tools

Why Use TheHive, Cortex, and Elasticsearch?

TheHive: An open-source Security Incident Response Platform (SIRP) used to create cases, assign tasks, and track progress of investigations.

Cortex: Works alongside TheHive to run analyzers on observables (like IPs, hashes) and return job summaries with threat intelligence.

Elasticsearch: Stores and indexes all case-related data to support fast searching, filtering, and reporting.

Together, they form an integrated, powerful, and scalable forensic investigation platform.

- Incident Tracking & Organization Centralized documentation ensures no step is missed during investigations.
- Collaboration Among Analysts Multiple users can work together with assigned tasks, improving SOC coordination.
- Efficient Evidence Analysis Automatically analyzes observables (IPs, files, hashes) using Cortex analyzers, saving time.
- Audit and Reporting Maintains case history and generates reports for compliance and better decisions.
- Scalability & Automation Easy to deploy and scale with Docker; supports automation while allowing manual control.

INTEGRATION & CONFIGURATION

```
thehive@thehive:~$ ls
thehive-cortex-docker
thehive@thehive:~$ cd thehive-cortex-docker/
thehive@thehive:~/thehive-cortex-docker$ ls
cortex  Cortex-Analyzers  docker-compose.yml  thehive
thehive@thehive:~/thehive-cortex-docker$ sudo docker-compose up -d
[sudo] password for thehive:
Starting cassandra    ... done
Starting elasticsearch ... done
Starting cortex       ... done
Starting thehive      ... done
thehive@thehive:~/thehive-cortex-docker$ _
```

- The command `docker-compose up -d` starts all services defined in the `docker-compose.yml` file.
- The `-d` (detached mode) flag runs the containers in the background.
- It indicates that the application stack has been launched successfully without blocking the terminal.

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```
thehive@thehive:~/thehive-cortex-docker$ sudo docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
69756fbabf54	strangebee/thehive:5	"/opt/thehive/entryp..."	7 seconds ago	Up 6 seconds	0.0.0.0:9000->9000/tcp, :::9000->9000/tcp
09efa0871fc4	thehiveproject/cortex:3.1.1	"/opt/cortex/entrypo..."	7 seconds ago	Up 6 seconds	0.0.0.0:9001->9001/tcp, :::9001->9001/tcp
0b64b9107e4d	docker.elastic.co/elasticsearch/elasticsearch:7.17.9	"/bin/tini -- /usr/l..."	8 seconds ago	Up 7 seconds	9200/tcp, 9300/tcp
985bf42644b9	cassandra:3.11	"docker-entrypoint.s..."	8 seconds ago	Up 7 seconds	7000-7001/tcp, 7199/tcp, 9042/tcp, 9160/tcp

- The docker ps output confirms successful deployment of all core components in the TheHive-Cortex environment.
- Containers for TheHive, Cortex, Elasticsearch, and Cassandra are running without errors.
- All services are in an “Up” status, indicating a stable and fully operational setup.

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```
GNU nano 7.2
version: '3.7'

services:
  cassandra:
    image: cassandra:3.11
    container_name: cassandra
    environment:
      - MAX_HEAP_SIZE=512M
      - HEAP_NEWSIZE=100M
    volumes:
      - cassandra_data:/var/lib/cassandra
    networks:
      - thehive

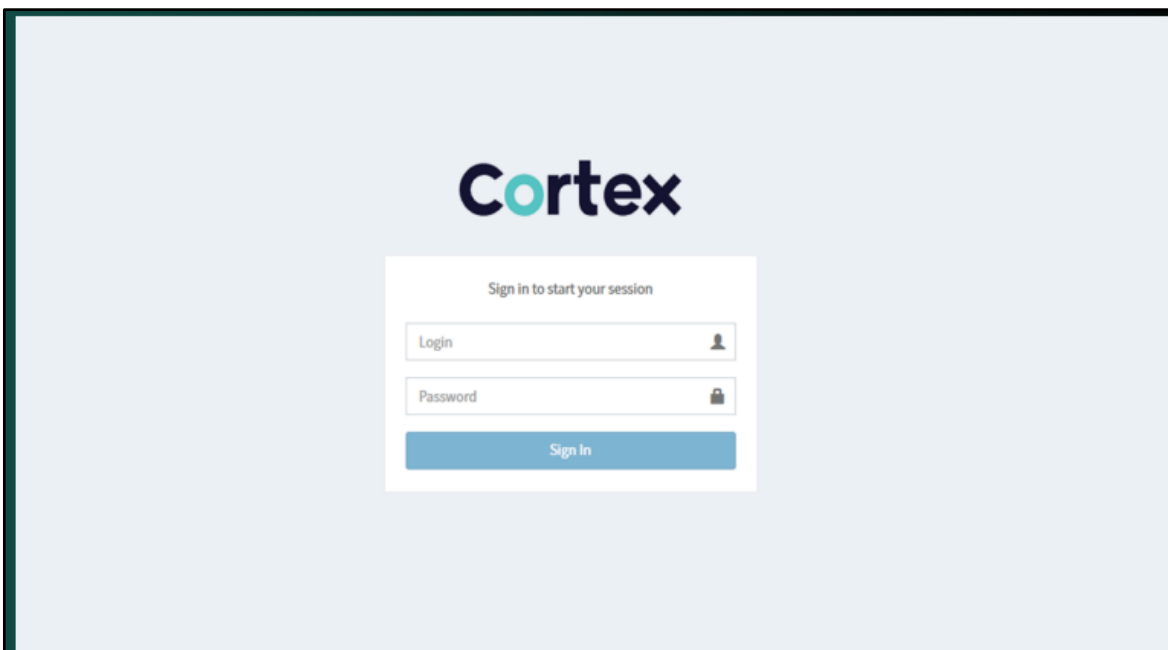
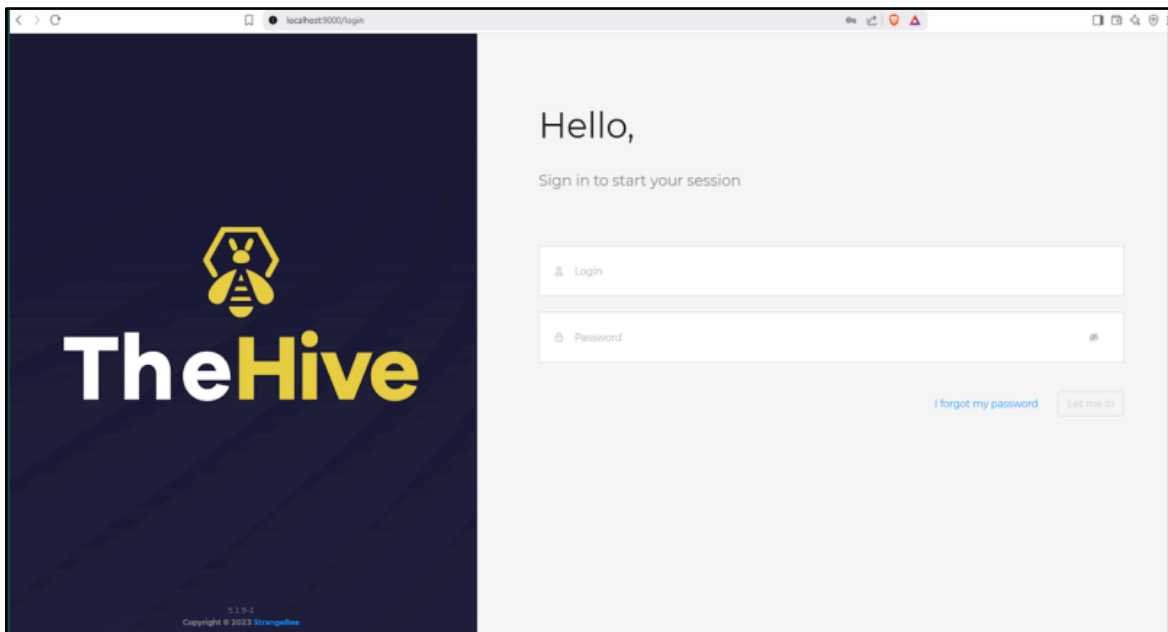
  elasticsearch:
    image: docker.elastic.co/elasticsearch/elasticsearch:7.17.9
    container_name: elasticsearch
    environment:
      - discovery.type=single-node
      - ES_JAVA_OPTS=-Xms1g -Xmx1g
    ulimits:
      memlock:
        soft: -1
        hard: -1
    mem_limit: 2g
    volumes:
      - esdata:/usr/share/elasticsearch/data
    networks:
      - thehive

  thehive:
    image: strangebee/thehive:5
    container_name: thehive
    depends_on:
      - cassandra
      - elasticsearch
    ports:
      - "9000:9000"
    volumes:
      - ./thehive/application.conf:/etc/thehive/application.conf
    networks:
      - thehive

  cortex:
    image: thehiveproject/cortex:3.1.1
    container_name: cortex
```

- This screenshot shows a dockercompose.yml configuration file for deploying the TheHive-Cortex stack using Docker Compose.
- It defines four services: Cassandra (database), Elasticsearch (search engine), TheHive (incident response platform), and Cortex (analyzer engine).
- The file specifies container images, memory settings, port mappings, volumes for data persistence, and service dependencies, ensuring proper communication and structured deployment of the environment

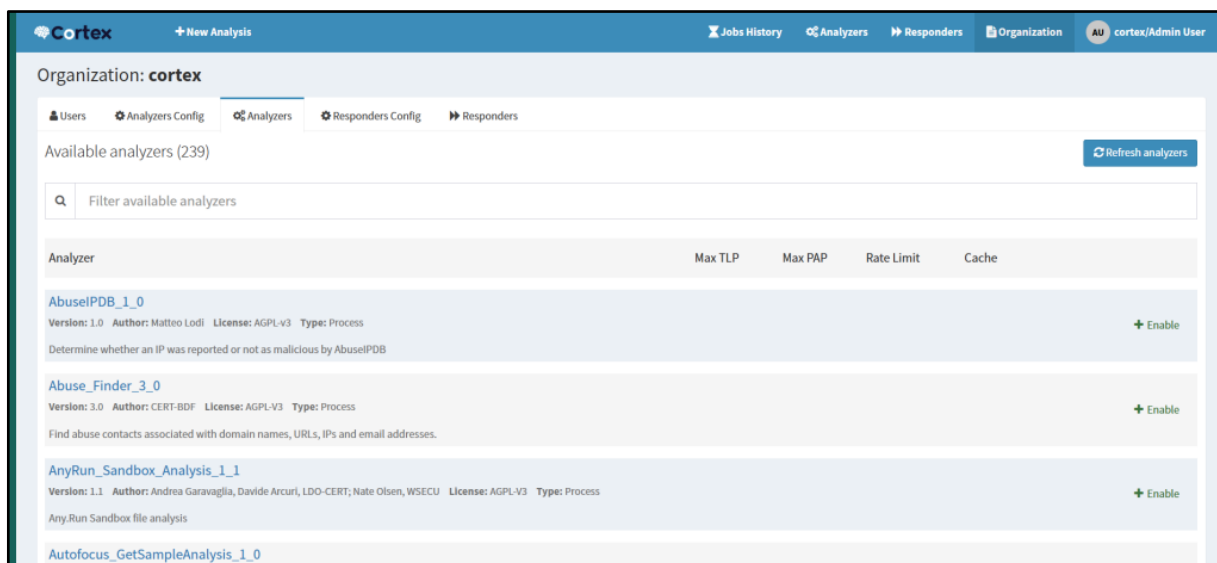
WEB INTERFACE ACCESS



Cloning and Configuring Cortex Analyzers via Terminall

```
thehive@thehive:~/thehive-cortex-docker$ cd Cortex-Analyzers/
thehive@thehive:~/thehive-cortex-docker/Cortex-Analyzers$ ls
analizers  AUTHORS  CHANGELOG.md  code_of_conduct.md  COMPONENTS  docs  images  jobs  LICENSE  README.md  responders  SECURITY.md  thehive-templates  utils
thehive@thehive:~/thehive-cortex-docker/Cortex-Analyzers$ cd analizers/
thehive@thehive:~/thehive-cortex-docker/Cortex-Analyzers/analizers$ ls
Abuse_Finder  CuckooSandbox  FileInfo  IPInfo  NSRL  Shodan  dnorella
AbuseIPDB  CyberChef  FireEyeISight  IPvoid  OSqueryLookup  SinkDB  unshortenlink
AvyRun  CyberCrime-Tracker  FireHOLBlockLists  IWE  Oryphe  SoltraEdge  URLhaus
AutoFocus  Cyberprotect  ForcepointWebSensePing  JoeSandbox  ONYXActiveScan  SophosIntelli  Uriscan.io
Baur  Cylance  FortiGuard  Jupyter_Analyzer  OpenCTI  SpamAssassin  Valhalla
BackscatterID  Glario  Gatewatcher_CTI  KasperskyTIP  OtxQuery  SpamhausBCL  ValidateObservable
BitcoinAbuse  DNSDB  GoogleDNS  LooqQuery  PaloAltoWildFire  Sploit  Verifalia
CIRP  HNSDumpster  Hooglestretiousing  Malpedia  PassiveTotal  StatusNetworks  VirusShare
Cms  DNSLookJugless  GoogleVisionAPI  Maltiverse  Patrol  StackSearch  VirusTotal
Censys  DNSSinkhole  GreyNoise  MalwareBeazar  PayloadSecurity  StopForumSpam  VMRay
CERTatPassiveDNS  DomainWallSPFOWARC  ORR  MalwareClustering  PhishingInitiative  TalosReputation  Vulners
CheckPhish  DomainTools  Hashkit  Malwares  PhishTank  TeamCyberWR  WOT
CICLHashlookup  DomainToolsIris  HIBP  MaxMind  ProofPoint  TestAnalyzer  Yara
CICLPassiveDNS  DShield  Hippocampe  MetDefender  Pulsedive  Threatcrowd  Yeti
CICLPassiveSSL  EchoTrail  HunterIO  NISP  QrDecode  ThreatGrid  Zscaler
CISCAP  EjecticID  Hdrjdnalysis  NISPWarningLists  RecordedFuture  ThreatMiner
ClashW  Elasticsearch  IBMXForce  MnemonicFENS  RiskIQ  ThreatResponse
Cluster25  EmailRep  Inlets  NGEntraID  Robtex  Thunderstorm
CrowdSec  EmergingThreats  IntegerCommunity  NspParsen  SecurityTrails  TorBtoMxle
CrowdStrikeFalcon  EmulParam  Investigate  NERO  SentinelOne  TorProject
Curlsh  Falc0nSandbox  TP-BPT  Nessus  SexIneIDRe  Tronix
```

Cloning and Configuring Cortex Analyzers Via Web-interface



TheHive & Cortex Integration Status



———— Versions ————

TheHive	5.1.9-1
---------	---------

———— Cortex ————

cortex0 - 3.1.1-1	OK
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———— MISP ————

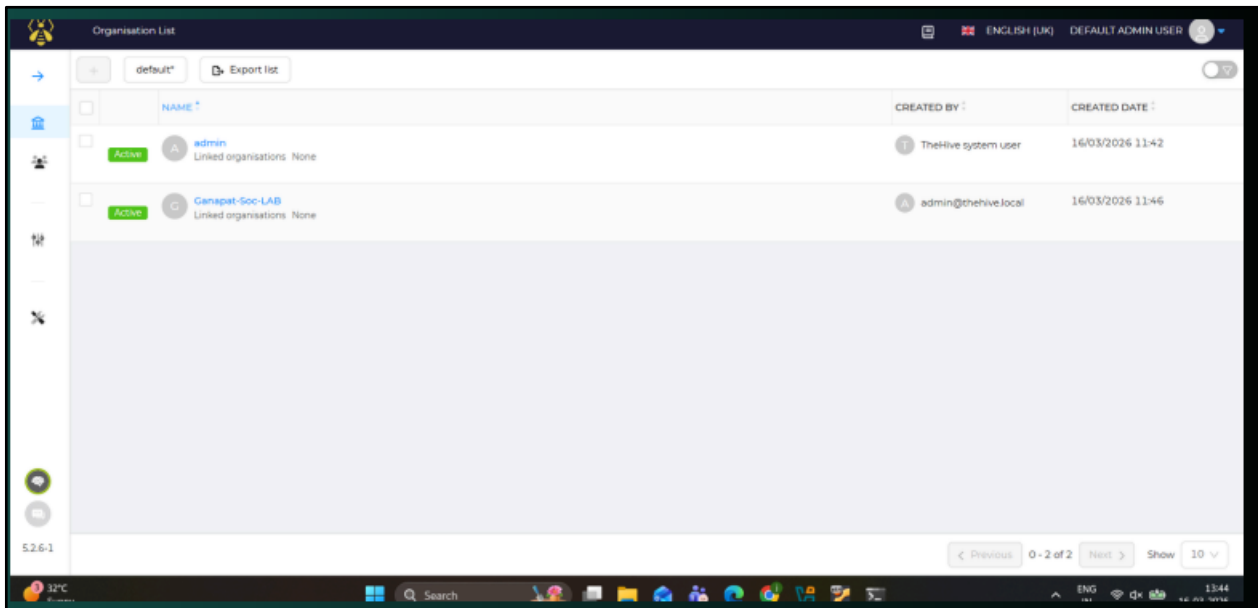
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———— License ————

License plan	Not Found
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- TheHive (v5.1.9-1) is successfully deployed and operational.
- Integration with Cortex (v3.1.1-1) is functioning correctly, as indicated by the “OK” status.
- The “License Plan: Not Found” confirms that the Community Edition is currently in use.

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- First, we create an organization in TheHive. Organization helps to manage users, alerts, cases, and investigations in a structured way. By creating a separate organization like Ganapat-Soc-LAB, the SOC team can work together and keep their data organized.
- It also helps to separate different teams or projects inside the same platform.

6. Proof of Concept

Adding a New User in TheHive

Type
Normal
Service users are essentially used for bots (API key authentication).

Organisation
Ganapat-Soc-LAB

* Login
std4@example.com

* Name
Student-1

* Profile
org-admin

Permissions
accessTheHiveFS manageAction manageAlert/create manageAlert/delete manageAlert/import
manageAlert/reopen manageAlert/update manageAnalyse manageCase/changeOwnership
manageCase/create manageCase/delete manageCase/merge manageCase/reopen manageCase/update
manageCaseReport manageCaseReportTemplate manageCaseTemplate manageComment
manageConfig manageCustomEvent manageFunction/create manageFunction/invoke
manageKnowledgeBase manageObservable managePage managePageTemplate manageProcedure

- The screenshot displays the “Adding a User” configuration window in TheHive, used to create a new user account within the organization.
- The administrator must define the user type, organization, login credentials, and assign an appropriate role-based profile (readonly, analyst, or org-admin).
- This step ensures proper role-based access control (RBAC) and secure user management within the platform.

Case Management And Analysis tool

Creating a New Case in TheHive

The screenshot displays the 'Create case' form in TheHive. The form is structured as follows:

- Title:** Malware Investigation - Hash Analysis
- Date:** 2025-04-17
- Severity:** LOW, MEDIUM, **HIGH**, CRITICAL
- TLP:** TLP-WHITE, TLP-GREEN, **TLP-AMBER**, TLP-RED
- PAP:** PAP-WHITE, PAP-GREEN, **PAP-AMBER**, PAP-RED
- Tags:** MalwareBazaar, SHA256, Hash, IOC...
- Description:**

This case involves the manual investigation of a suspected malicious file identified by its SHA256 hash. The hash was submitted for verification using the MalwareBazaar analyzer.

Initial suspicion indicates potential malware activity linked to file-sharing, botnet activity, or dropper behavior. Further analysis is required to determine its origin, behavior, and threat level.
- Tasks:** Custom fields, Add a task
- Footer:** Cancel, Confirm

- The screenshot shows the “Create Case” interface in TheHive, used to initiate a new incident investigation.
- The case is configured with a defined severity level (High), TLP and PAP classification (Amber), relevant tags, and a detailed investigation description.
- This structured case creation process ensures proper incident documentation, classification, and workflow management within the SOC environment.

Case Management And Analysis tool

Adding and Assigning Tasks to a Case

The screenshot shows the 'Adding a Task' modal in TheHive. It contains the following fields and controls:

- Group:** A dropdown menu with 'default' selected.
- Title:** A text input field containing 'Use Cortex + MalwareBazaar analyzer to check reputation and family info of the given hash'.
- Mandatory:** A toggle switch that is currently turned off, with the text 'At least one log must be present' below it.
- Description:** A rich text editor with the text 'find malware'.
- Assignee:** A dropdown menu with 'cortex@example.com' selected.
- Flag this task?:** A toggle switch that is currently turned on.
- Due date:** A date and time picker showing '2025-04-19 20:51:55'.
- Buttons:** 'Cancel', 'Save and add another', and 'Confirm' at the bottom right.

- The screenshot displays the “Adding a Task” interface in TheHive, used to assign specific investigative actions within a case.
- The task is configured to utilize the MalwareBazaar analyzer via Cortex to validate the reputation and malware family of a given hash.
- It includes task ownership, due date, and priority flagging to ensure structured workflow tracking and accountability within the incident response process.

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Adding Observable to a Case

The screenshot displays the 'Adding an Observable' dialog box in TheHive. The interface is organized into several sections: 'Type' is set to 'hash'; 'Value' contains a long alphanumeric string (SHA256 hash); 'TLP' (Traffic Light Protocol) is set to 'TLP-AMBER'; 'PAP' (Privacy and Accountability Policy) is set to 'PAP-AMBER'; 'Is IOC' (Indicator of Compromise) is a toggle switch turned off; 'Has been sighted' is a toggle switch turned off; 'Ignore similarity' is a toggle switch turned off; 'Tags' is an empty text field; and 'Description' is a rich text editor with a preview button. At the bottom, there are three buttons: 'Cancel', 'Save and add another', and 'Confirm'.

- The screenshot shows the “Adding an Observable” interface in TheHive, where an indicator (SHA256 hash) is being added to an active case.
- The observable is classified with appropriate TLP and PAP levels (Amber), ensuring controlled information sharing and sensitivity handling.
- This step enables structured IOC management and allows integration with analyzers such as Cortex for automated threat intelligence enrichment.

Case Management And Analysis tool

default

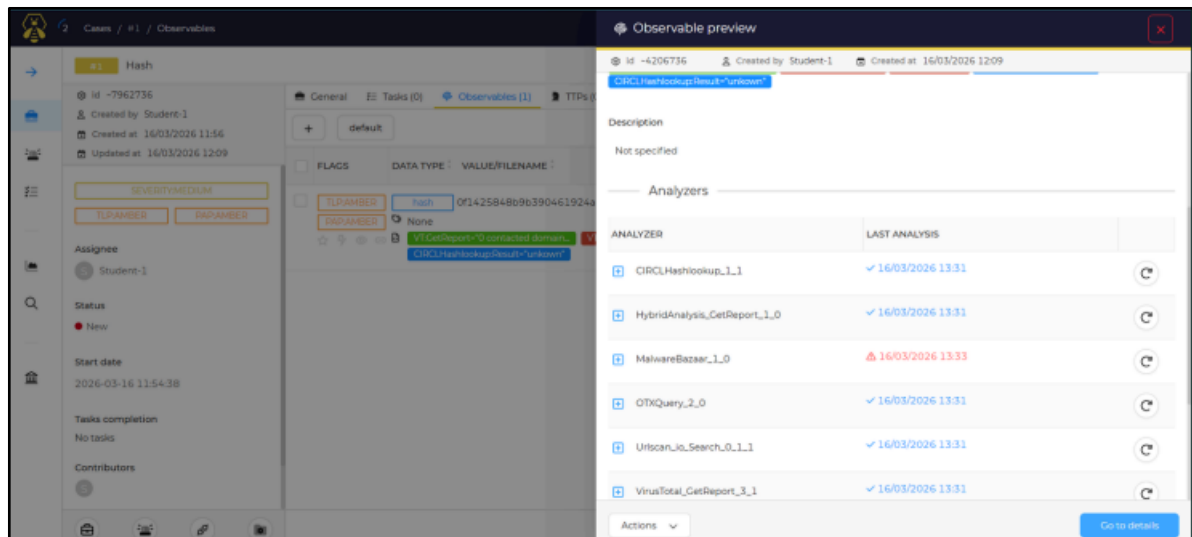
Quick Filters

Export list

</

- After creating the organization and users, we created cases in TheHive. In each case, we added observables such as Hash, IP address, and URL for investigation.
- These cases help SOC analysts to track and analyze suspicious activities. The case list shows important details like severity, observables, tasks, and investigation status, which helps analysts manage and monitor security incidents properly.

Case Management And Analysis tool



- After adding observables in the case, we run analyzers to investigate the data. The analyzers check the observables like IP address, URL, or Hash using different threat intelligence sources.
- This helps the SOC analyst to know whether the data is malicious or safe. The analyzer results provide useful information such as reputation, threat score, and related security details, which helps in making the final investigation decision.

Case Management And Analysis tool

Hash File

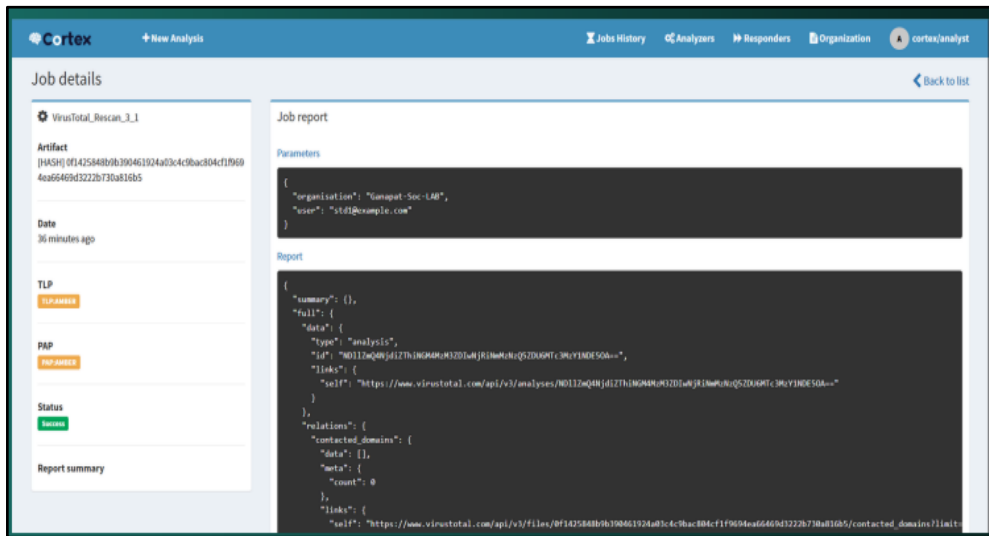
0f1425848b9b390461924a03c4c9bac804cf1f9694ea66469d3222b730a816b5

Analyzer Results

Success	[hash] 0f1425848b9b390461924a03c4c9bac804cf1f9694ea66469d3222b730a816b5	Analyzer: CIRCLHashlookup_1_1	Date: 29 minutes ago	User: cortex/analyst	TLP:AMBI:B	TLP:AMBI:B	View	Delete
Success	[hash] 0f1425848b9b390461924a03c4c9bac804cf1f9694ea66469d3222b730a816b5	Analyzer: HybridAnalysis_GetReport_3_0	Date: 29 minutes ago	User: cortex/analyst	TLP:AMBI:B	TLP:AMBI:B	View	Delete
Success	[hash] 0f1425848b9b390461924a03c4c9bac804cf1f9694ea66469d3222b730a816b5	Analyzer: OTXQuery_2_0	Date: 29 minutes ago	User: cortex/analyst	TLP:AMBI:B	TLP:AMBI:B	View	Delete
Success	[hash] 0f1425848b9b390461924a03c4c9bac804cf1f9694ea66469d3222b730a816b5	Analyzer: Urlscan_io_Search_0_1_1	Date: 29 minutes ago	User: cortex/analyst	TLP:AMBI:B	TLP:AMBI:B	View	Delete
Success	[hash] 0f1425848b9b390461924a03c4c9bac804cf1f9694ea66469d3222b730a816b5	Analyzer: VirusTotal_GetReport_1_1	Date: 29 minutes ago	User: cortex/analyst	TLP:AMBI:B	TLP:AMBI:B	View	Delete
Success	[hash] 0f1425848b9b390461924a03c4c9bac804cf1f9694ea66469d3222b730a816b5	Analyzer: VirusTotal_Rescan_3_1	Date: 29 minutes ago	User: cortex/analyst	TLP:AMBI:B	TLP:AMBI:B	View	Delete

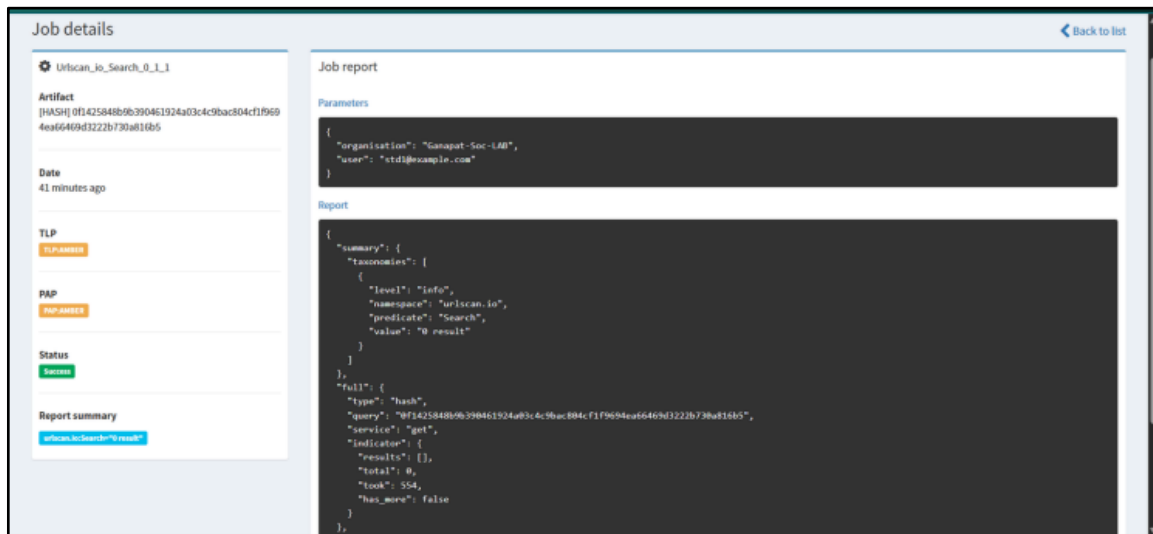
- After adding the hash observable, we ran different analyzers in Cortex such as VirusTotal, OTX Query, Hybrid Analysis, CIRCL Hash Lookup, and URLScan.
- These analyzers check the hash using different threat intelligence platforms.
- The results help the SOC analyst to understand whether the file is malicious, suspicious, or safe. All analyzer results are shown in the case, which helps in better investigation and decision making.

VirusTotal-Analyzer Report



- In this step, we check the observable using the VirusTotal analyzer in Cortex.
- The analyzer sends the hash to VirusTotal and collects the analysis report. The report shows detailed information about the file, such as reputation, related domains, and threat intelligence data.
- This helps the SOC analyst understand whether the file is malicious or safe.

URLScan Analyzer Result



- In this step, we used the URLScan analyzer in Cortex to check the observable. URLScan searches the observable in its database to find any related malicious activity or suspicious information.
- In this result, no malicious activity was found for the hash, which means the observable appears safe in the URLScan database.

Case Management And Analysis tool

IP Address

23.26.129.38

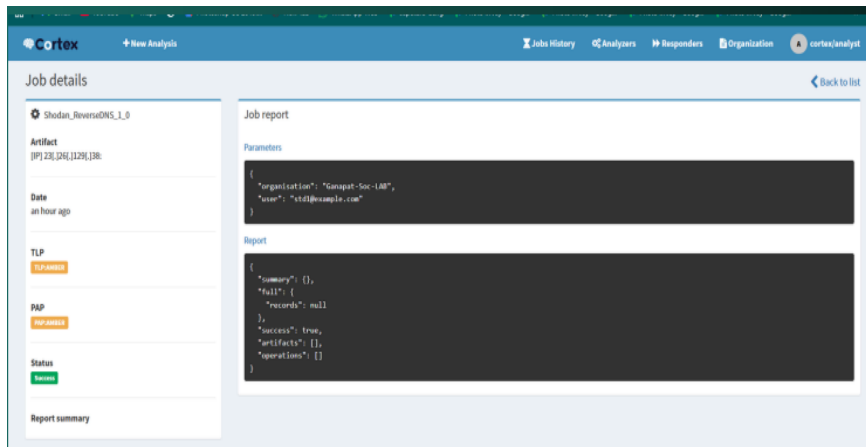
Analyzer Results

Status	Job details	TLP	PAP
Success	[ip] 23[.]26[.]129[.]38: Analyzer: VirusTotal_GetReport_3_1 Date: an hour ago User: cortex/analyst	TLP:AMBER	PAP:AMBER View Delete
Success	[ip] 23[.]26[.]129[.]38: Analyzer: Shodan_ReverseDNS_1_0 Date: an hour ago User: cortex/analyst	TLP:AMBER	PAP:AMBER View Delete
Success	[ip] 23[.]26[.]129[.]38: Analyzer: OTXQuery_2_0 Date: an hour ago User: cortex/analyst	TLP:AMBER	PAP:AMBER View Delete
Success	[ip] 23[.]26[.]129[.]38: Analyzer: GoogleDNS_resolve_1_0_0 Date: an hour ago User: cortex/analyst	TLP:AMBER	PAP:AMBER View Delete

- In this step, we analyzed the IP address observable using different analyzers in Cortex.
- Tools like VirusTotal, Shodan Reverse DNS, OTX Query, and Google DNS were used to collect information about the IP address.
- These analyzers help the SOC analyst identify if the IP is malicious, suspicious, or safe by checking threat intelligence databases.

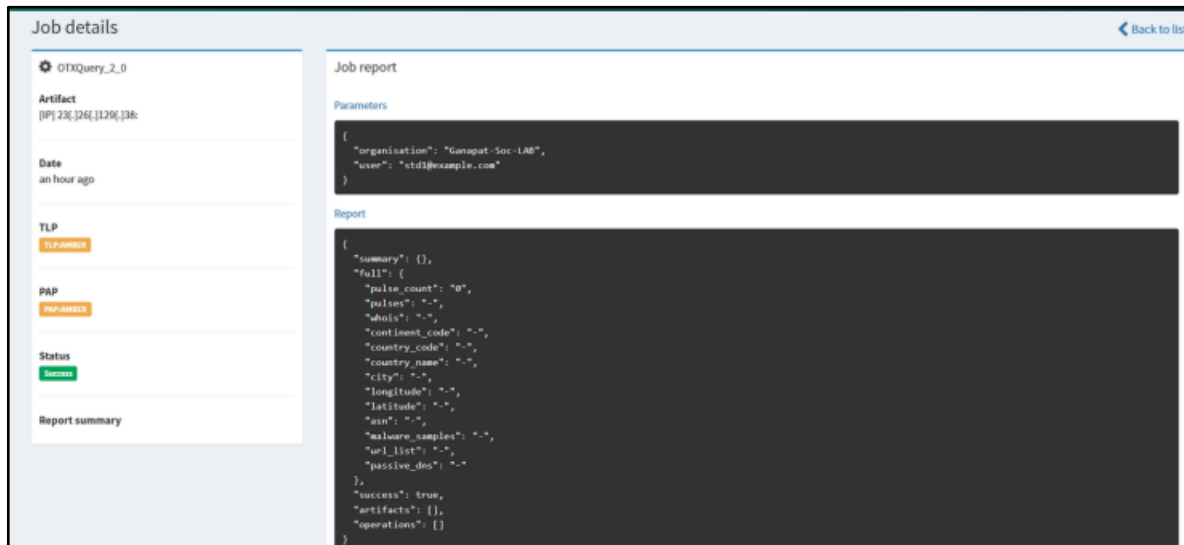
Case Management And Analysis tool

Shodan Reverse DNS Analysis



- In this step, we analyzed the IP address using the Shodan Reverse DNS analyzer in Cortex.
- This analyzer checks the IP address in the Shodan database to find related domain information and network details. The result helps the SOC analyst understand whether the IP address is associated with any suspicious activity.

OTX Threat Intelligence Analysis



- In this step, we analyzed the IP address using the OTX Query analyzer in Cortex.
- OTX (Open Threat Exchange) is a threat intelligence platform that helps identify malicious indicators such as IP addresses, URLs, and file hashes.
- The analyzer checks the IP address in the OTX database to find any related threat information. This helps SOC analysts understand if the IP address is associated with any known cyber threats.

URL

xxps://urlhaus[.]abuse[.]ch/url/319532/

Case Management And Analysis tool

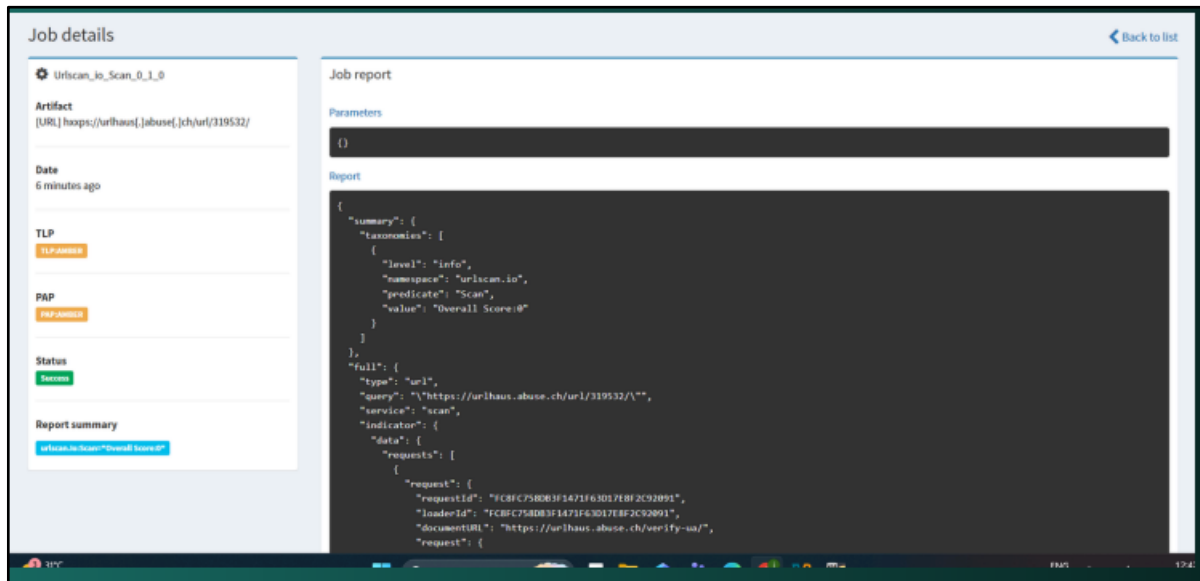
Analyzer Results

Success	[url] https://urlhaus[.]abuse[.]ch/url/319532/	Analyzer: urlscan_io_Scan_0_1_0	Date: 3 minutes ago	User: cortex/analyst	TLP:AMBER	DWP:AMBER	View	Delete
Success	[url] https://urlhaus[.]abuse[.]ch/url/319532/	Analyzer: GoogleSafebrowsing_2_0	Date: 14 minutes ago	User: cortex/analyst	TLP:AMBER	DWP:AMBER	View	Delete
Success	[url] https://urlhaus[.]abuse[.]ch/url/319532/	Analyzer: VirusTotal_GetReport_3_1	Date: 43 minutes ago	User: cortex/analyst	TLP:AMBER	DWP:AMBER	View	Delete
Success	[url] https://urlhaus[.]abuse[.]ch/url/319532/	Analyzer: VirusTotal_Scan_3_1	Date: 43 minutes ago	User: cortex/analyst	TLP:AMBER	DWP:AMBER	View	Delete
Success	[url] https://urlhaus[.]abuse[.]ch/url/319532/	Analyzer: OTXQuery_2_0	Date: 44 minutes ago	User: cortex/analyst	TLP:AMBER	DWP:AMBER	View	Delete
Success	[url] https://urlhaus[.]abuse[.]ch/url/319532/	Analyzer: HybridAnalysis_GetReport_1_0	Date: 44 minutes ago	User: cortex/analyst	TLP:AMBER	DWP:AMBER	View	Delete

- In this step, we analyzed the URL observable using multiple analyzers in Cortex such as URLScan, Google Safe Browsing, VirusTotal, OTX Query, and Hybrid Analysis.
- These tools check the URL in different threat intelligence platforms to detect any malicious or phishing activity.
- This helps the SOC analyst understand whether the URL is safe or harmful.

Case Management And Analysis tool

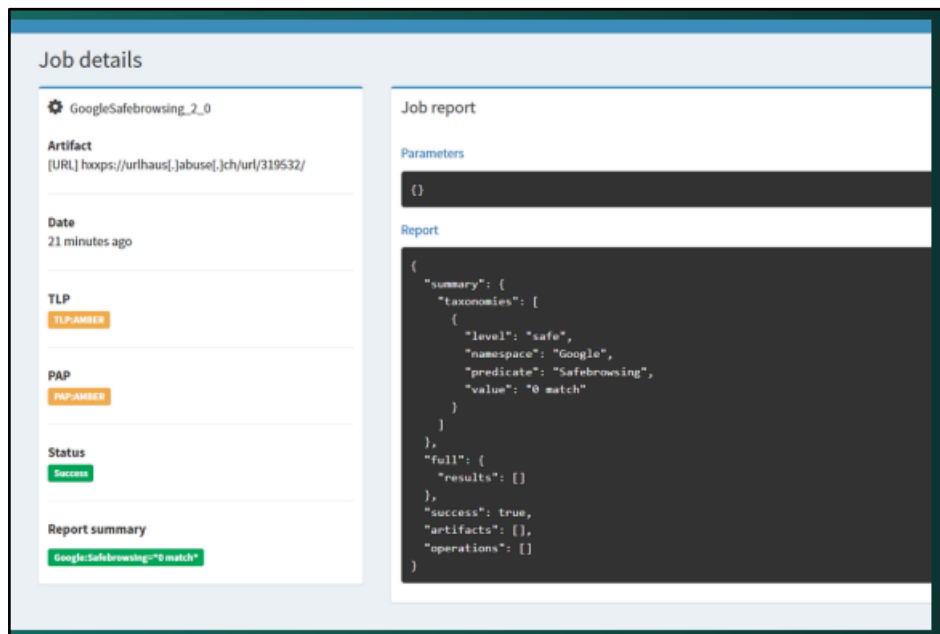
URLScan Detailed Report



- In this step, we performed a detailed scan of the URL using the URLScan analyzer in Cortex.
- The analyzer checks the URL and provides an overall score based on its behavior and activity.
- In this result, the overall score is 0, which means no suspicious or malicious activity was detected. This helps the SOC analyst confirm that the URL is safe.

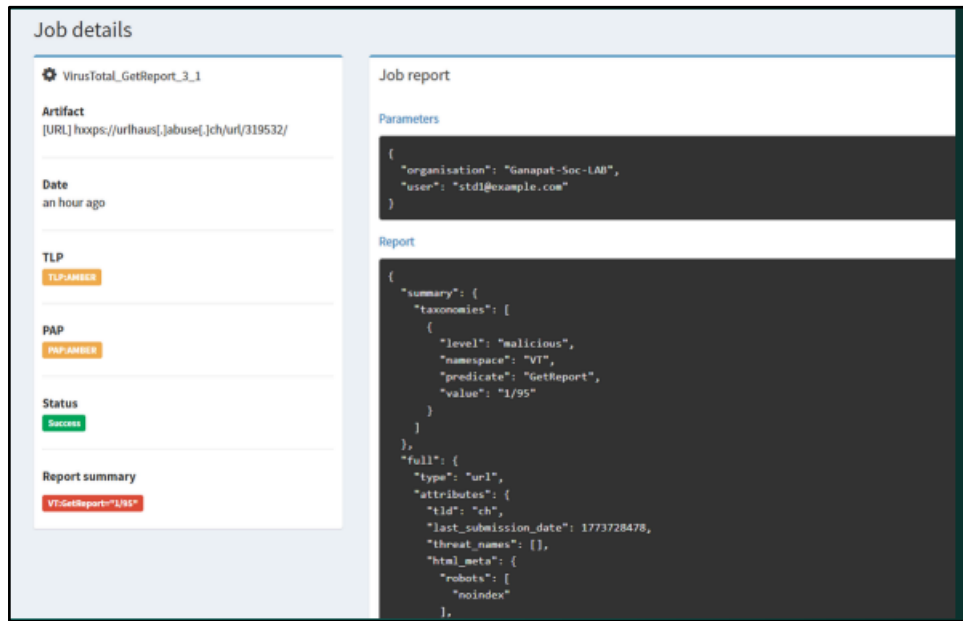
Case Management And Analysis tool

Google Safe Browsing Analysis



- In this step, we checked the URL using Google Safe Browsing analyzer in Cortex.
- This tool verifies whether the URL is listed as malicious, phishing, or unsafe.
- In this result, it shows “No match” which means the URL is not detected as harmful and is considered safe.

VirusTotal URL Analysis



- In this step, we analyzed the URL using VirusTotal analyzer in Cortex.
- VirusTotal checks the URL across multiple security engines. In this result, 1 out of 95 engines detected the URL as malicious, which means the URL may be suspicious but not strongly confirmed as malicious.
- This helps the SOC analyst take a careful decision during investigation.

File

[file] 073933625bf9f2f7b3c93d43ccbdef7c8500af30cf43a1e5e0537760760d8b1f[.]zip

Case Management And Analysis tool

Analyzer Results

Jobs History (21)

Data Types (8) Job Type (2) Analyzers (18) Observable

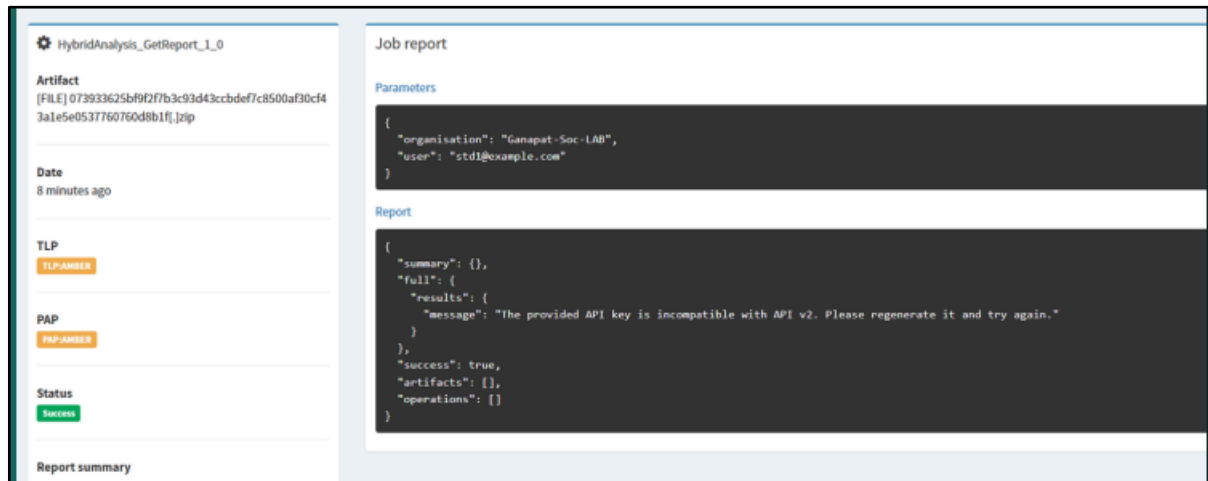
Select - Select - Select - Search for observable data Search Clear

Pagination 50 / page

Status	Job details	TLP	PAP
Success	[file] 073933625b9f2f7b3c93d43ccbdef7c8500af30cf43a1e5e0537760760d8b1f.zip Analyzer: HybridAnalysis_GetReport_1_0 Date: 3 minutes ago User: cortex/analyst	TLP:AMBER	PAP:AMBER View Delete
Success	[file] 073933625b9f2f7b3c93d43ccbdef7c8500af30cf43a1e5e0537760760d8b1f.zip Analyzer: VirusTotal_Scan_3_1 Date: 3 minutes ago User: cortex/analyst	TLP:AMBER	PAP:AMBER View Delete
Success	[file] 073933625b9f2f7b3c93d43ccbdef7c8500af30cf43a1e5e0537760760d8b1f.zip Analyzer: VirusTotal_GetReport_3_1 Date: 3 minutes ago User: cortex/analyst	TLP:AMBER	PAP:AMBER View Delete
Success	[file] 073933625b9f2f7b3c93d43ccbdef7c8500af30cf43a1e5e0537760760d8b1f.zip Analyzer: OTXQuery_2_0 Date: 3 minutes ago User: cortex/analyst	TLP:AMBER	PAP:AMBER View Delete

- In this step, we analyzed a suspicious file using multiple analyzers in Cortex such as VirusTotal, Hybrid Analysis, and OTX.
- The file is scanned to check if it contains any malware or harmful content.
- These analyzers provide detailed information about the file's behavior and reputation. This helps the SOC analyst determine whether the file is safe or malicious.

Hybrid Analysis Result



- In this step, we tried to analyze the file using the Hybrid Analysis analyzer in Cortex.
- However, the result shows an error message indicating that the API key is incompatible.
- This means the analyzer could not fetch the report due to a configuration issue. Such errors can occur if the API key is outdated or not properly configured.

YARA File Analysis

Yara_3_0

Artifact
[FILE] 073933625bf9f2f7b3c93d43ccbdef7c8500af30cf43a1e5e0537760760d8b1f.zip

Date
a few seconds ago

TLP
TLP-AMBER

PAP
PAP-AMBER

Status
Success

Report summary
Yara:Match="0/1 rule(s)"

Job report

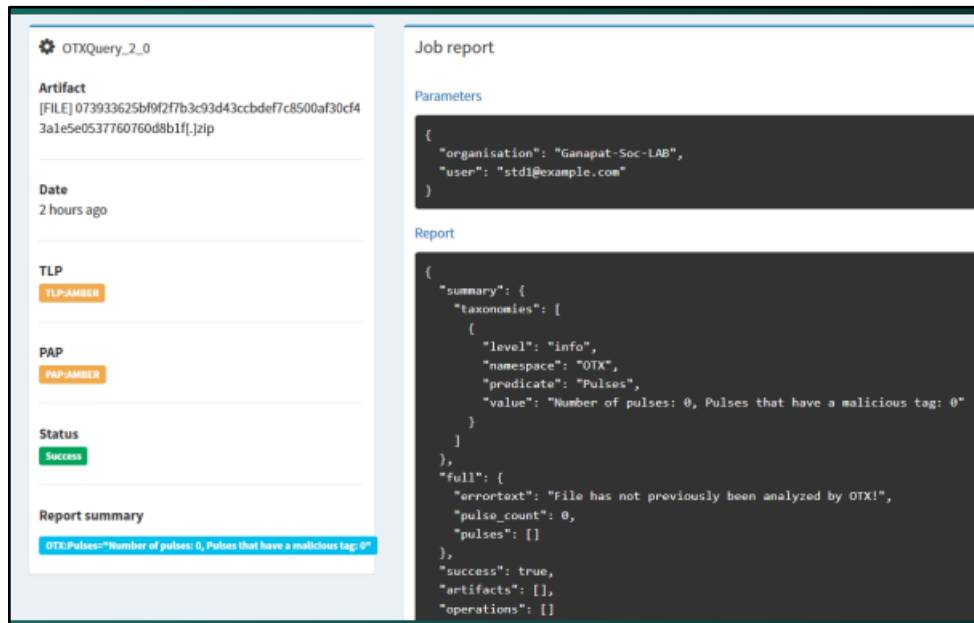
Parameters
{}

Report

```
{
  "summary": {
    "taxonomies": [
      {
        "level": "safe",
        "namespace": "Yara",
        "predicate": "Match",
        "value": "0/1 rule(s)"
      }
    ]
  },
  "full": {
    "results": [],
    "summary": {
      "taxonomies": [
        {
          "level": "safe",
          "namespace": "Yara",
          "predicate": "Match",
          "value": "0/1 rule(s)"
        }
      ]
    }
  }
}
```

- In this step, we analyzed the file using the YARA analyzer in Cortex.
- YARA is used to detect malware patterns based on predefined rules.
- In this result, it shows 0 out of 1 rules matched, which means no malicious pattern was detected in the file. This indicates that the file appears safe based on YARA analysis.

OTX File Analysis



- In this step, we analyzed the file using the OTX Query analyzer in Cortex.
- OTX checks whether the file is associated with any known threats.
- In this result, it shows 0 pulses and no malicious activity, which means the file has not been reported as malicious in the OTX database. This indicates that the file appears safe based on OTX analysis.

Domain

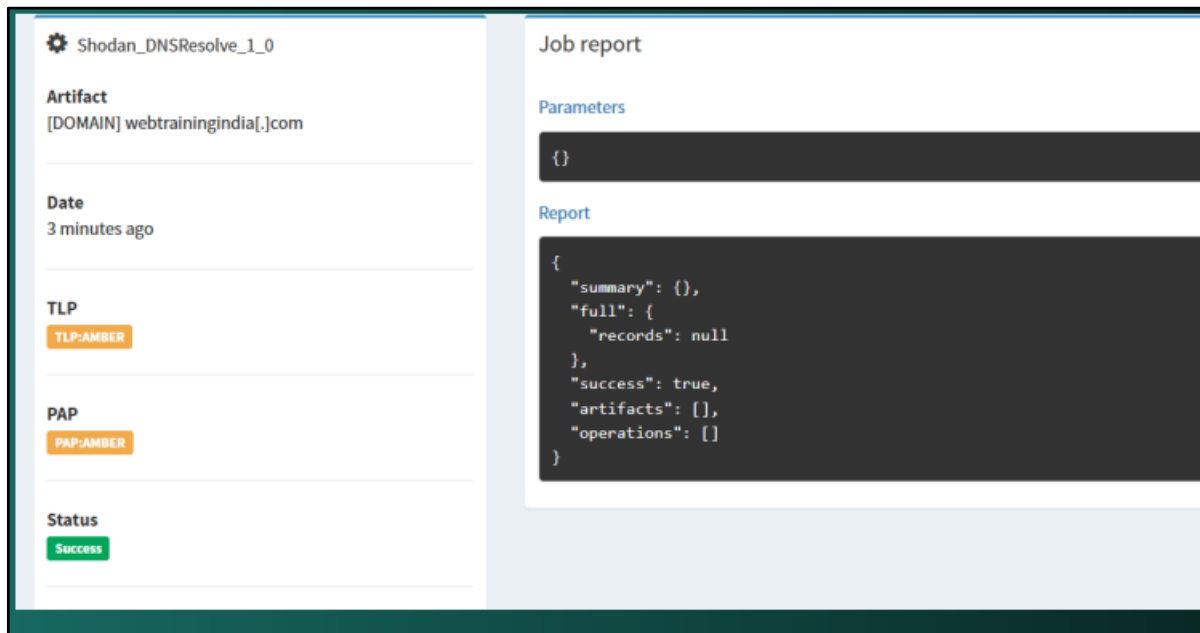
webtrainingindia.com

Case Management And Analysis tool

Status	Job details	TLP	PAP
Success	[domain] webtrainingindia[.]com Analyzer: Shodan_DNSResolve_1_0 Date: a few seconds ago User: cortex/analyst	TLP:AMBER	PAP:AMBER
Success	[domain] webtrainingindia[.]com Analyzer: VirusTotal_GetReport_3_1 Date: 37 minutes ago User: cortex/analyst	TLP:AMBER	PAP:AMBER
Success	[domain] webtrainingindia[.]com Analyzer: Urlscan_io_Scan_0_1_0 Date: 37 minutes ago User: cortex/analyst	TLP:AMBER	PAP:AMBER
Success	[domain] webtrainingindia[.]com Analyzer: Shodan_DNSResolve_1_0 Date: 37 minutes ago User: cortex/analyst	TLP:AMBER	PAP:AMBER
Success	[domain] webtrainingindia[.]com Analyzer: OTXQuery_2_0 Date: 37 minutes ago User: cortex/analyst	TLP:AMBER	PAP:AMBER
Success	[domain] webtrainingindia[.]com Analyzer: HybridAnalysis_GetReport_3_0 Date: 37 minutes ago User: cortex/analyst	TLP:AMBER	PAP:AMBER

- Cortex such as VirusTotal, URLScan, Shodan DNS Resolve, OTX Query, and Hybrid Analysis.
- These tools check the domain across different threat intelligence platforms to identify any malicious activity.
- This helps the SOC analyst understand whether the domain is safe or associated with any threats.

Shodan DNS Analysis



- In this step, we analyzed the domain using the Shodan DNS Resolve analyzer in Cortex.
- This analyzer checks the domain to find related DNS and network information.
- In this result, no records were found, which means there is no significant data available for this domain in the Shodan database. This indicates that the domain does not show any suspicious activity from this source.

Case Management And Analysis tool

URLScan Domain Analysis

The screenshot displays the URLScan Domain Analysis interface. The left sidebar contains the following information:

- Artifact:** [DOMAIN] webtrainingindia[.]com
- Date:** 44 minutes ago
- TLP:** TLP:AMBER
- PAP:** PAP:AMBER
- Status:** Success
- Report summary:** urlscan.io:Scan:"Overall Score:0"

The main panel shows the job report with the following parameters and report:

Parameters

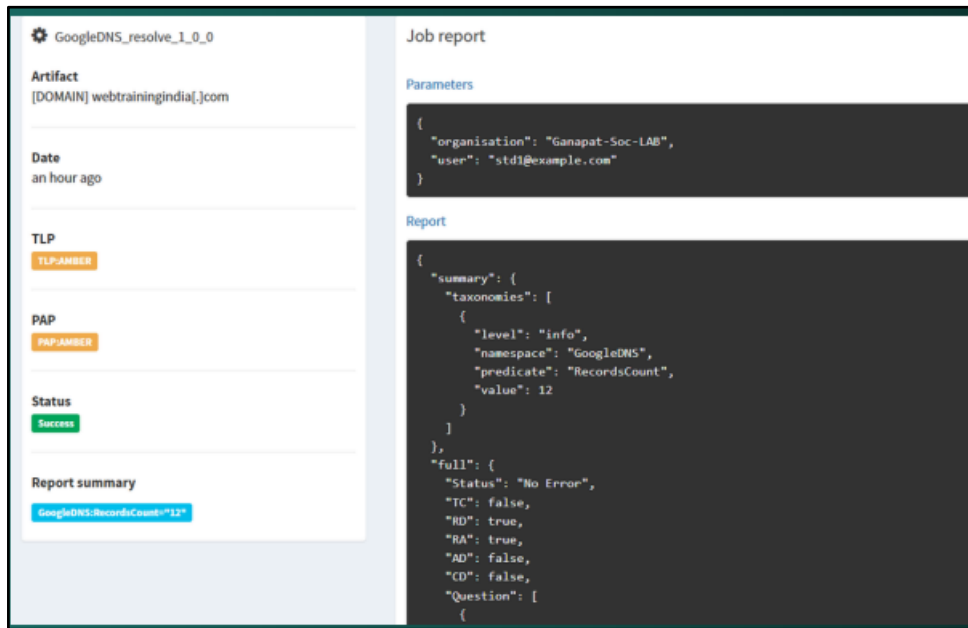
```
{
  "organisation": "Ganapat-Soc-LAB",
  "user": "std1@example.com"
}
```

Report

```
{
  "summary": {
    "taxonomies": [
      {
        "level": "info",
        "namespace": "urlscan.io",
        "predicate": "Scan",
        "value": "Overall Score:0"
      }
    ]
  },
  "full": {
    "type": "domain",
    "query": "\"webtrainingindia.com\"",
    "service": "scan",
    "indicator": {
      "data": {
        "requests": [
```

- In this step, we analyzed the domain using the URLScan analyzer in Cortex.
- URLScan checks the domain's behavior and activity on the web.
- In this result, the overall score is 0, which means no suspicious or malicious activity was detected. This indicates that the domain appears safe based on URLScan analysis.

Google DNS Analysis



- In this step, we analyzed the domain using the Google DNS Resolve analyzer in Cortex.
- This analyzer checks the DNS records of the domain.
- In this result, it shows 12 DNS records found, which means the domain is active and has proper DNS configuration. This helps the SOC analyst understand the domain's network presence.

7. Future Scope

The **Case Management and Analysis Tool** provide an efficient system for handling and analyzing cases; however, there is significant scope for further enhancement and adaptation to modern technological needs. The following improvements can be considered for future development:

1. Integration of Artificial Intelligence (AI) and Machine Learning (ML)

Future versions of the system can incorporate AI and ML techniques to improve automation and decision-making.

- Predictive Analysis: ML models can predict case outcomes, priority levels, and resolution time based on historical data.
- Automated Case Classification: AI can automatically categorize and assign cases to the appropriate departments.
- Anomaly Detection: Detect unusual patterns or high-risk cases for faster response.

2. Cloud-Based Deployment

The system can be deployed on cloud platforms such as AWS, Azure, or Google Cloud.

- Enables remote access and real-time collaboration.
- Provides scalability for handling large volumes of cases.
- Ensures secure data storage with backup and recovery features.

3. Real-Time Notification and Alert System

The tool can be enhanced with real-time communication features.

- Instant notifications for case updates, assignments, and deadlines.
- Email/SMS alerts for critical or high-priority cases.
- Integration with messaging platforms for faster communication.

4. Mobile Application Support

Developing a mobile version of the system will improve accessibility.

- Users can manage and track cases anytime, anywhere.
- Push notifications for instant updates.
- Improved user engagement and productivity.

5. Advanced Reporting and Dashboard

Future enhancements can include more advanced analytics.

- Interactive dashboards with graphs and visual reports.
- Performance tracking of users and departments.
- Data-driven insights for better decision-making.

6. Integration with Other Systems

The tool can be integrated with external systems for better functionality.

- Integration with CRM, ERP, or IT service management tools.
- API-based connectivity for data exchange.
- Seamless workflow across multiple platforms.

8. References

1. [□ https://www.google.com](https://www.google.com)
2. [□ https://www.techtarget.com](https://www.techtarget.com)
3. [□ https://www.geeksforgeeks.org](https://www.geeksforgeeks.org)
4. [□ https://www.thegeekstuff.com](https://www.thegeekstuff.com)
5. [□ https://www.ibm.com/topics/case-management](https://www.ibm.com/topics/case-management)
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7. [□ https://www.atlassian.com/itsm/service-request-management](https://www.atlassian.com/itsm/service-request-management)
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10. [□ https://www.servicenow.com/products/itsm/what-is-case-management.html](https://www.servicenow.com/products/itsm/what-is-case-management.html)